

Nirasha Aluthgamage

Industrial Automation & Robotics Undergraduate | Manufacturing, Process & Automation Engineering

Colombo, Sri Lanka | +94 77 423 5364 | jayaninirasha18@gmail.com | LinkedIn: [Nirasha Aluthgamage](#)

PROFILE

Engineering Technology undergraduate specializing in Industrial Automation and Robotics at the University of Kelaniya (four years completed, expected graduation in 2027). Practical exposure across beverage and ice-cream manufacturing, and glove manufacturing, with contributions to PLC-based automation systems, water recovery infrastructure, and ISO 50001:2018 energy auditing. Skilled in PLC programming, Lean Six Sigma tools, and CAD design (SolidWorks, AutoCAD). Seeking internship, placement, or junior executive opportunities in manufacturing, process, or automation engineering.

SKILLS

Automation & Controls: PLC Programming (TIA Portal, Siemens, Ladder Logic), HMI Programming, Motor Controlling, Hydraulic & Pneumatic Systems, CNC Programming (G-code)

Design & Engineering Software: SolidWorks, AutoCAD, Altium Designer, FluidSIM, RoboDK, MATLAB, ProfiCAD

Manufacturing & Process Improvement: Lean Six Sigma, DMAIC, OEE Tracking, Root Cause Analysis (RCA), Value Stream Mapping (VSM), Total Productive Maintenance (TPM), 5S Implementation, SOP Development

Quality & Safety: ISO 50001:2018 Energy Audit, Safety Health & Environment (SHE) Standards, Occupational Health & Safety (OH&S)

Programming: Python, C Programming

Other: MS Office Suite (Word, Excel, PowerPoint, Outlook)

EDUCATION

Bachelor of Engineering Technology (Hons) Degree in Industrial Automation and Robotics

2022 - 2026

Faculty of Computing & Technology | University of Kelaniya, Sri Lanka

WORK EXPERIENCES

Electrical & Electronic Engineering Intern

August 2025 – March 2026

Ceylon Cold Stores PLC (Elephant House) | Ranala, Sri Lanka

- Contributed to several industrial projects throughout the research, design and factory floor installation.
- Prepared the Bill of Quantities (BOQ) and co-authored a Request for Proposal (RFP) for vendor bidding
- Gained hands-on exposure to automated Glass Line & PET Line bottling operations and the ice cream production process
- Participated in the Annual Factory Maintenance & Overhaul and safety toolbox meetings

Semi-Skilled Worker

March 2021 – August 2022

ATG® - Intelligent Glove Solutions | EPZ Wathupitiwala, Sri Lanka

- Operated and monitored glove manufacturing machines across continuous shift operations, following SOPs and 5S standards to maintain consistent quality output
- Performed routine quality checks and collaborated with maintenance and quality teams to resolve machine faults and process deviations
- Maintained workplace cleanliness and safety standards throughout tenure

PROJECTS

Industrial Projects | Ceylon Cold Stores PLC (Elephant House), Sri Lanka

Design of an Automated Final Rinse Water Recovery System for CIP system 2025

- Designed an automated 10,000 L CIP final rinse water recovery system and automated divert valve network, driving resource conservation, reducing industrial wastewater effluent, and improving utility cost efficiency.

Automated Line Monitoring and Dry-Run Protection System for Vat Tank Changeover 2025

- Designed a PLC-based dual-zone indicator system with stack lights and alarms to prevent pump dry-run damage during syrup vat changeovers.

Design & Development of a Conveyor Merge Traffic Control System 2025 - 2026

- Contributed to the project by building a sensor-driven, relay-based interlocking system to eliminate bottle collisions and line jams at a high-speed conveyor junction, from schematic design to on-floor installation.

ISO 50001:2018 Energy Audit | 2026

- Served as an internal auditor documenting a Non-Conformities Report (NCR) covering energy wastage across the factory.

Electrical RCD Test 2026

- Performed a factory-wide RCD (ELCB) Test using Kyoritsu 5406A Digital RCD Tester ensuring electrical safety.

Temperature-based Valve Control System for the Aging Room 2026

- Designed an automated solenoid valve and RTD sensor-based system to safely divert hot CIP washdown water away from the Ice Banks, replacing manual drain valve operation.

Design and Optimization of a Conveyor Segment to Facilitate Warmer Relocation and Eliminate Bottlenecks at Soft Drink Factory 2026

- Redesigned the existing conveyor layout, prepared the Bill of Quantities (BOQ), and authored a Request for Proposal (RFP) for vendor bidding.

Academic & Capstone Research Projects | University of Kelaniya, Sri Lanka

Automated Feeding System 2024

- Designed an automated animal feeding system using ESP32 and Firebase to manage surplus food sustainably, with sensors detecting food levels and a Firebase-powered mobile app enabling real-time control and monitoring. The system was developed using Arduino IDE for hardware programming and Firebase for backend integration and data management.

Manually Operated Pneumatic Robotic Arm for Flat-Pack Box Handling and Folding Operations 2024

- The project focused on the design and development of a manually operated pneumatic robotic arm integrated with a conveyor belt system, which can lift the flat-pack boxes and fold the caps of the boxes, enhancing efficiency in packaging operations. The design process was carried out using FluidSIM software.

Design and Fabrication of a PID – Based Liquid Level Control System 2025

- Designed and implemented a PID-based system using Arduino, ultrasonic sensing, and DC pumps for automated liquid level control with real-time LCD monitoring and stable setpoint regulation.

Designed and simulated a robotic pick-and-place work cell in RoboDK 2025

- Used two UR20 arms with RobotiQ 2F-85 grippers to safely transfer fragile glass bottles from a table to a conveyor, and then into a crate, with precise motion planning and safety measures.

Design & Development of an Autonomous Roboshuttle System for Warehouse Inventory Transport with Machine Learning-Enabled Scheduling & Monitoring 2026

- Developed an autonomous differential-drive AMR with synchronized dual lead screw vertical lifting and electromagnetic gripping. Integrated AprilTag monocular vision localization and edge-based linear regression scheduling.

PUBLICATIONS

Abstract publication under the project titled Automated Feeding System | University of Kelaniya **2024**

- The Research Center of the Faculty of Computing and Technology (RC-FCT), together with ICATC 2024, is organized the 4th Undergraduate Research Symposium of the Faculty of Computing and Technology, University of Kelaniya. [The 4th Undergraduate Research Symposium 2024 Proceedings](#)

CONFERENCES & RESEARCH PARTICIPATION

Workshop Participant: 'Designing IoT Solution Using a Single-Board Computer' | 9th International Conference on Advances in Technology and Computing (ICATC)

Faculty of Computing & Technology | University of Kelaniya, Sri Lanka | December 2024

Conference Participant: 2nd International Multidisciplinary Desk Research Conference (DRC 2024)

Faculty of Computing & Technology | University of Kelaniya, Sri Lanka | October 2024

Presenter & Project Exhibitor: 4th Undergraduate Research Symposium

Faculty of Computing & Technology | University of Kelaniya, Sri Lanka | November 2024

- Abstract Presentation: Authored and presented an academic abstract detailing the engineering and design methodologies of an automated mechanism.
- Project Exhibition: Selected to showcase the prototype project 'Automated Feeding System' to a panel of academic judges and industry professionals.

VOLUNTEER EXPERIENCES

Member of Engineering Technology Students' Association **2022 – Present**

- Member of Media and Editorial Team

Member of Green Club of FCT **2023 – 2026**

- Organizing committee member of Sustainable Mobility Workshop powered by John Keels Holdings

Member of IEEE IES Students Chapter UOK **2024 - 2026**

- Member of Designing & Editorial Team
- Organizing committee member of Distinguish Lecturer Program

LANGUAGES

Sinhala - Native or bilingual proficiency

English - Professional working proficiency

REFERENCES

Dr. Tharaga Sharmilan,

Senior Lecturer,

Department of Applied Computing,

Faculty of Computing and Technology,

University of Kelaniya.

tharagas@kln.ac.lk - 0778329675

Mr. Malith Ranathunga,

Senior Manager - Electrical & Electronic Engineering,

Ceylon Cold Stores PLC, Ranala, Sri Lanka.

MalithThushan.ccs@keells.com - 0771763702